

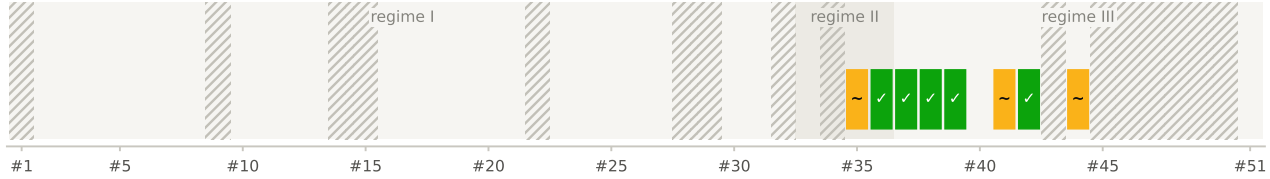
H118: The forked RPG is two replicas of one dynamics: damage spreading between rooms (NE15, #35)

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 2/5 Usefulness 1.5/5

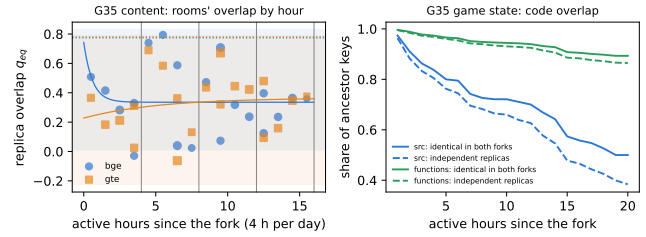
Bottom line. After the NE15 fork the rooms' content shared no same-day state beyond day 1 and was far from identical replicas from the first hour, while their code diverged slowly with a frozen core and correlated damage; the content damage time and any rules plateau are not identified at these counts.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH358 treats the NE15 fork as a replica experiment. On 2026-03-16 the village split into #best (3 agents) and #rest (9), and each room took its own copy of the RPG built in #34: the same rules and starting state, different players. In kinetic Ising, two replicas of one dynamics either stay together (frozen or field-ordered phase) or separate at a rate set by the noise (damage spreading). H118 measures the overlap of the two rooms' content per active hour and the overlap of the two forks' code per hour. **Content and code behave oppositely.** The rooms' content is already far from identical replicas in the first hour and shares no same-day state after day 1. The code separates slowly on the edit clock, keeps a frozen core of untouched files, and its damage is correlated.



Observables. Left: hourly replica overlap of the #35 rooms (bge filled, gte squares; size = statement weight), with the fitted relaxation, the identical-replica level from joint relabels (dotted) and the identical-kickoff band (shaded). The overlap is low from the first hour and flat over five days. Right: the share of ancestor src files and functions identical in both forks (solid) against independent replicas (dashed). Code damage grows over 20 active hours and stays correlated.

PHYSICAL MODEL

Model 02 (replicas of a kinetic spin dynamics) with model-11 content spins. Content: $\mathbf{c}_r(b)$ is room r 's agent-weighted centroid of style-residualized statement vectors in active hour b , centred on a leave-own-period-out regime reference so that the shared static component stays in. Split halves give unbiased squared norms S_r . Code: each ancestor key (file, function) is a spin that keeps or changes its inherited value; two independent replicas are identical at a key with probability c_{ACB} .

$$q(b, b') = \frac{\mathbf{c}_A(b) \cdot \mathbf{c}_B(b')}{\sqrt{\langle S_A \rangle \langle S_B \rangle}}, \quad M(d) = q_{eq}(d) - q_{lag}(d), \quad q_{eq}(t) = q_\infty + (q_0 - q_\infty)e^{-t/\tau_D}$$

$$X = q_{code} - c_{ACB}, \quad F = P(\text{unchanged in both forks})$$

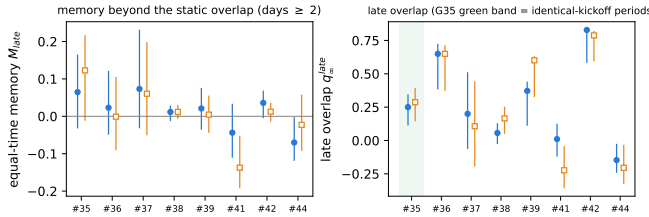
SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator who forks a team onto copies of one artifact: the copies' conversations stop sharing a state within the first hour, so expect no residual coordination from the common start. The artifacts themselves diverge slowly and at the same hot spots: about half of the inherited code stays untouched in both copies, and identity exceeds independent copies by 0.12. A shared artifact acts as a field on where work lands, not on what the teams say. Two numbers to plan with: content overlap at the first hour 0.51 (vs 0.80 for random teams), code identity after five days 0.50.

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RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
P0 synthetic: memory size, τ_D , plateau, drive	memory test fixed to M_{late} (size 0.07–0.13); τ_D slow-world power 0.58; plateau PI power 0.16–0.29; drive caught 100%	1/4 as written (Amendment 1)
P1 G35 fast: $\tau_D \leq 4$ h, no memory	τ_D 0.6 h, CI 0.25–80 (gte 5.2 h, no decay); M_{late} 0.07 [−0.03, 0.17]	memory met; τ_D inconclusive
P2 no plateau; HH kill	q_{∞}^{late} 0.25 [0.11, 0.35], band rank 2/5 (36a 0.62); kill silent	consistent, unpowered
P3 code slow, frozen, $X > 0$	day 1 0.84 (pred. ≥ 0.85); end 0.50; F 0.50; X 0.116, p 6e-12	3/4
P4 replication: no memory $\geq 5/7$	6/7 in each model; $\tau_D \leq 4$ h	3/7 memory met
P5 control (G38, G44 below band median)	G38, G44 0.06, −0.15 (gte 0.17, −0.21)	supported
HH358	content fast-looking but unidentified; code slow and correlated	mixed



Second figure. Left: equal-time memory M_{late} (days ≥ 2) per two-room period with agent $\times$ bin bootstrap 95% intervals (bge filled, gte open). It covers 0 in #35 and 6/7 kickoff restarts; #44 (and #41 in gte) is negative: same-day anti-alignment. Right: late overlap q_{∞}^{late} . The two different-kickoff periods (#38, #44) are the lowest, as the control predicts; #35 sits inside the identical-kickoff band.

GOAL PERIODS IN DETAIL

G35 $\sim$ mixed
G36 $\checkmark$ supported
G37 $\checkmark$ supported
G38 $\checkmark$ supported
G39 $\checkmark$ supported
G41 $\sim$ mixed
G42 $\checkmark$ supported
G44 $\sim$ mixed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

One fork event with 3 #best agents: agent-bootstrap intervals are wide, and in regime II no agent constants exist, so composition is part of the day-1 separation. The real-skeleton synthetic shows that τ_D and the plateau rule are not identifiable at these counts, so “no plateau” is not a negative result. Memory in this estimator sees only same-day shared state; slow memory shared across days is invisible to it. Code identity is keyed (moved code counts as changed), and the game-state predictions were written after reading H07’s day-level results.

ROUND 2

Hierarchical τ_D across the eight kickoffs; agent constants for regime-II agents from #33 and #36a; test whether #34 touch counts (H07’s C1) predict the shared hot spots behind X ; link same-day anti-alignment to room-specific operator messages. Confirmation on #45–#47 and #50 (frozen, not run).

- **Essence.** whether a shared artifact acts as a field on where forked teams work (code hot spots) while their conversations decorrelate at once.
- **H118-R1** Hierarchical damage time: pool the hourly relaxation across the eight kickoffs with period-specific τ_D and shrinkage (CLAUDE.md exception (d)); one period cannot identify it.
- **H118-R2** Composition-free overlap: agent constants for regime-II agents from #33 and #36a remove R-comp from G35’s day-1 separation.
- **H118-R3** Where damage lands: test whether H07’s #34 touch counts predict the shared hot spots behind the identity excess X (H07’s C1).
- **H118-R4** Same-day anti-alignment (G41, G44): link negative memory to room-specific operator messages per day.

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